The GTE Polynomial as Unified Field Theory: One 19-Bit Description for Spatial Dynamics, Gauge Coupling, Gravity, Entanglement, and Baryon Number

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Abstract

A single polynomial $p(L, C, R) = C + R - CR - LCR \pmod 7$, requiring $K_{\mathrm{CMCA}} = 19$ bits to specify, simultaneously generates five fundamental physical structures: Rule 110 Turing universality ($\mathbf{A}_{\mathrm{Lean}}$), Standard Model gauge vertex conservation via $\mathbb{Z}_7$ winding arithmetic ($\mathbf{A}_{\mathrm{Lean}}$), gravitational coupling via the Poisson equation $\nabla^2\Phi = G_{\mathrm{eff}}\, p(w_x, w_y, w_z)$ ($\mathbf{A/D}$), quantum entanglement with CHSH parameter $S = 2.4459$ (86.5% of the Tsirelson bound, $\mathbf{A/D}$), and the Born rule from Page–Wootters analysis ($\mathbf{A/D}$). The additional specification cost for each role beyond the first is exactly zero.

From the same 19-bit description applied to the three-tape architecture ([7]): (i) color confinement as an MDL theorem (description-length gap $\Delta K \approx 3.17$ bits); (ii) $N_c = 3$ from the tape count alone; (iii) baryon number $B = \frac{1}{3}\sum_j \chi_q(w_j)$ as a topological charge; (iv) SM fermion/boson split from the non-primitive roots of $\mathbb{Z}_7^*$; and (v) lepton-W universality from shared $\mathbb{Z}_7$ winding sectors.

The PMDL variational principle yields gravity as local MDL minimization; the halting-weighted MDL residual from PSC undecidability gives the cosmological constant, whose census-capacity evaluation is the zero-parameter value $\Omega_\Lambda = 0.6899$ ($+0.18\sigma$ from Planck 2018). Gravity and the Λ floor are the record and measure sectors of one Gaussian generating functional $\ln Z[J]$. The vacuum $w = 0$ is the unique fixed point of $p(x, x, x) \equiv x \pmod 7$, establishing vacuum stability from first principles. All foundational results are machine-certified in Lean 4 with zero `sorry`.

Contents

What this paper claims and does not claim

Lean-certified ($\mathbf{A}_{\text{Lean}}$, zero `sorry`):

- [**Single-Source Principle**] The same 19-bit polynomial unifies five physical domains at $K_{\text{extra}} = 0$ each (spatial dynamics, gauge coupling, gravity, entanglement, baryon number) (`gte_polynomial_five_roles_k_extra_zero`, `FiveRolesPolynomial.lean`, zero sorry).
- [**MDL Uniqueness**] p is the unique cubic polynomial over GF(7) matching the mass hierarchy $\{0 \to 0,\ 2 \to 6,\ 3 \to 5,\ 4 \to 5,\ 6 \to 5\}$ with $K_{\text{extra}} = 0$ (`unique_cubic_gravity_coupling`).
- [**Vacuum Fixed Point**] $w = 0$ is the unique fixed point of $p(x, x, x) \equiv x \pmod 7$; 5 is not a quadratic residue mod 7 (`vacuum_unique_fixed_point_z7`).
- [**Charge**] $3Q(w) = p(0, w, 0) = w$ for all PSC winding w; EM is the degree-1 center projection of p (`gte_charge_is_linear_projection`).
- [**Gravity/EM Split**] Gravity $= p(w, w, w)$ (degree 3); EM $= p(0, w, 0) = w$ (degree 1); both from 19 bits (`gravity_em_degree_split`).
- [**SU(3) Gluons**] 8 gluon charge vectors from $\Delta w = \pm 1$ constraint (`su3_gluon_charge_vectors`, `su3_cmca_master_bundle`).
- [**Color Confinement**] $\Delta K = \log_2(9) \approx 3.17$ bits MDL cost of free colored quarks forbids them as asymptotic states (`color_confinement_k_extra_pos`).
- [**Baryon Number**] $B = \frac{1}{3} \sum_j \chi_q(w_j)$ is a topological charge conserved by the Bianchi identity; $B = 1/3$ per quark from $N_{\text{tapes}} = 3$ (`gte_baryon_number_topological_charge`).
- [**Fermion/Boson**] Sectors $\{2, 4, 6\}$ are the non-primitive roots of $\mathbb{Z}_7^*$; sector $\{3\}$ is the unique primitive root (`gte_winding_identifies_charged_fermions`).
- [**SRRG Bridge**] $1/\phi = (\sqrt{5} - 1)/2$ is the positive root of $p(x, x, x) = x$; equals the Self-Referential Renormalization Group (SRRG) contraction eigenvalue (`gte_poly_srrg_bridge`).
- [**Dark Energy**] $D_{\text{res}} > 0$ from PSC halting undecidability (`no_final_self_theory`, `psc_undecidability_residual_pos`, `d_res_determines_omega_lambda`, `psp_epoch_selection_master`).

Analytically derived, computationally confirmed (A/D):

- $Z[J]$ unified generating functional; all massless propagators $(-\Delta)^{-1} = 1/(4\pi r)$ (derived, not assumed).
- PMDL-Λ duality: gravity = the record sector, and the Λ floor = the measure sector, of the single Gaussian generating functional $\ln Z[J]$; the split $K_{\text{self}} = K_{\text{carrier}} + K_{\text{records}}$ is an exact complete-the-square identity following from $K_{\text{extra}} = 0$.
- D_{res} is the realized halting-weighted record ledger (limit-computable, not computable); its census-capacity evaluation $\Omega_\Lambda = (\ln 2/3\pi) \log_2(2000/3) = 0.6899$ ($+0.18\sigma$ from Planck 2018; zero free parameters) is the upper bracket, paired with the carrier floor $3\pi/14 = 0.67320$.
- $W^\pm$ as angular mode: $\Delta w = 3$, $m_W = g\, v_{\text{PSC}}/2 = 80.37$ GeV (0.003% from PDG 2024); $SU(2)_L$ gauging from PMDL ($\mathbf{A}_{\text{Lean}}$; zero named axioms; `su2l_l2_from_phimdl_potential_catad`).
- [**Exact force law**] $\varphi(b) = G_{\text{eff}} M \operatorname{erf}(b/\sqrt{2}\sigma_{\text{AL}})/(4\pi b)$; $F(b) \to G_{\text{eff}} M/(4\pi b^2)$ as $b \gg \sigma_{\text{AL}}$. Three equivalent mechanism levels: gradient kick (Level 1), linearized GR (Level 2), equivalence principle (Level 3).
- [**Born rule**] $P(k|\tau) = |\langle k|U(\tau)|\psi_0\rangle|^2$ derived from τ_c Page–Wootters clock; τ_c satisfies all PW prerequisites.

Computationally verified (A):

- All massless propagators equal the same $(-\Delta)^{-1}$; 3D CA rule impossibility; Bell $S = 2.4459$ (LHV excluded, 86.5% Tsirelson bound).

This paper does *not* claim:

- Full derivation of the quantum mechanism behind Ω_Λ (analytical derivation is complete; the quantum-geometric step is outlined as open).
- Baryon number at the QCD-sector level (string tension requires Level 2).

Claim-strength taxonomy

$$\mathbf{A}_{\text{Lean}} > \mathbf{A}_{\text{MDL}} > \mathbf{A/D} > \mathbf{A} > \mathbf{B} > \mathbf{D}$$

$\mathbf{A}_{\text{Lean}}$: machine-checked in Lean 4, zero `sorry`. *Conditional* $\mathbf{A}_{\text{Lean}}$ uses a named physics axiom (always stated explicitly); zero-sorry holds in the proof body. One named axiom remains open: `spinor_exchange_equals_2pi_rotation` (Level 3 braid atlas completion).
$\mathbf{A}_{\text{MDL}}$: MDL-unique within a declared expression class.
A/D: full analytic derivation; computationally confirmed.
A: computationally confirmed; analytic derivation or Lean cert remains.
B: all formula components GTE-derived; one structural mechanism not yet formally proved.
D: exploratory / long-term open.

The Two Laws: One Framework, Two Levels

Level 1 — The Algebraic Certificate (the discrete law, 19 bits):

$$\boxed{p(L,C,R) \;=\; C + R - CR - LCR \pmod 7}$$

A single four-term polynomial over GF(7). Certifies which structures the physical field must contain. Machine-certified unique ($\mathbf{A}_{\text{Lean}}$; P40, P48).

Level 2 — The Field Law (the continuum law):

$$\boxed{\mathcal{L}_{\Phi_{\text{MDL}}} \;=\; \tfrac{1}{2}(\partial_\mu \Phi)^2 \;-\; \frac{m^2}{49}\big(1-\cos 7\Phi\big) \;+\; \mathcal{L}_{\text{gauge}} \;+\; \mathcal{L}_\chi}$$

The unique Lorentz-invariant $\mathbb{Z}_7$-symmetric field realising the Level 1 certificate ($\mathbf{A}_{\text{Lean}}$; P42, P48).

The bridge — self-consistency forces one mass scale:

$$\boxed{p(x,x,x) = x \;\Longrightarrow\; x^* = \tfrac{1}{\varphi} \;\Longrightarrow\; m_\varphi = m_\tau}$$

The Level 2 field law is derived in P42 [4]. The complete two-level derivation: P48 [2].

1 Introduction

The standard model of particle physics encodes three gauge interactions through independent group-theoretic choices: U(1) for electromagnetism (EM), $SU(2)_L$ for the weak force, and SU(3) for color. Gravity is accommodated separately by general relativity, and quantum statistics—the assignment of fermionic or bosonic character to particle species—is a further independent postulate. The principal challenge of unification is to replace these independent inputs with a single organizing principle whose content is smaller than the sum of its parts. This paper is part of the Universal Generative Principle (UGP) Physics programme, the quantitative branch of the Reflexive Reality programme [6].

The Minimum Description Length (MDL) principle offers a precise formulation of "smaller than the sum of its parts": a unified description is one whose specification length is strictly less than the combined lengths of the parts described. A cellular automaton (CA) rule r of specification length K_r bits achieves MDL unification of n physical roles if the description length of all n roles together does not exceed $K_r + 0 \cdot n$, i.e., each additional role costs zero extra bits.

The central result. This paper establishes that the polynomial

$$p(L, C, R) \; = \; C + R - CR - LCR \pmod{7},$$

specified in $K_{\text{CMCA}} = 19$ bits, achieves MDL unification of five independent physical domains with $K_{\text{extra}} = 0$ for each:

(1) **Spatial dynamics:** p restricted to GF(2) inputs gives Rule 110, the unique Class 4 binary cellular automaton achieving Turing universality (Cook's theorem [1]).

(2) **Gauge coupling:** all 33 Standard Model (SM) charged-current vertices conserve $\mathbb{Z}_7$ winding arithmetic under p.

(3) **Gravity:** $\delta S_{\text{PMDL}}/\delta\Phi = 0$ gives $\nabla^2\Phi = G_{\text{eff}}\, p(w_x, w_y, w_z)$.

(4) **Quantum entanglement:** the cross-tape coupling generates entanglement negativity $\mathcal{N} = 0.24$ and Bell nonlocality $S = 2.4459$ (86.5% Tsirelson bound; LHV excluded).

(5) **Baryon number:** $B = \frac{1}{3}\sum_j \chi_q(w_j)$ is a topological charge conserved by the Bianchi identity.

These five roles are not features grafted onto the polynomial; they are algebraic consequences of the same 19-bit object. A polynomial that achieves any one of them achieves all five, as established by the Lean 4 machine-certified theorems enumerated in the claims box above and in Appendix A.

Architecture. This paper is the second of a two-paper series. The companion paper [7] establishes the three-tape Generative Triple Evolution (GTE) architecture: three chiral Minkowski cellular automata (CMCA tapes) sharing a common causal period τ_c, together with the algebraic lifting theorem that carries CA-level identities to the Φ_{MDL} continuum field. The present paper takes that architecture as given and derives its unification consequences.

The polynomial as certificate. The three-tape CMCA is the minimal algebraic specification of the Φ_{MDL} continuum field (established in [4,5]). The polynomial p is the kernel of that specification—the single compact rule from which all five physical roles are derived. When a role is said to be "generated" by the polynomial, we mean: the polynomial's algebraic properties are sufficient to certify that the corresponding structure exists in Φ_{MDL}, with zero additional specification cost. We do not mean that the polynomial dynamically produces physical particles, gravitons, or quarks. The certificate structure is detailed in the companion paper [7] §2.

Two distinct $\mathbb{Z}_7$-symmetric objects. The theory contains two distinct objects that both encode $\mathbb{Z}_7$ symmetry but play *different physical roles*:

- $p(L, C, R) = C + R - CR - LCR$ over GF(7) — the *update operator and coupling source*. It governs CA dynamics (spatial update rule), appears as the PMDL gravity source ($\nabla^2\Phi = G_{\text{eff}}\, p(w_x, w_y, w_z)$), and is the coupling at gauge vertices ($\mathbb{Z}_7$ winding conservation).
- $V_{\mathbb{Z}_7}(\Phi) = (m^2/49)(1 - \cos 7\Phi)$ — the *potential energy*. It appears in the Φ_{MDL} Lagrangian $\mathcal{L} = \frac{1}{2}(\partial\Phi)^2 - V_{\mathbb{Z}_7}$, stabilizes BPS kink solutions (via the BPS condition $\partial_x\Phi = \sqrt{2V_{\mathbb{Z}_7}}$), and defines the $\mathbb{Z}_7$ vacuum structure at $\Phi_k = 2\pi k/7$.

Both are $\mathbb{Z}_7$-periodic and vanish at the $\mathbb{Z}_7$ vacua; neither replaces the other. The continuum extension $p_{\text{cont}}(\phi_x, \phi_y, \phi_z) = \phi_z + \phi_y - \phi_y\phi_z - \phi_x\phi_y\phi_z$ (a real-valued polynomial with the same algebraic form as p) serves as the gravity source in the Level-2 PMDL equation. See `gap_g1_polynomial_continuum_taxonomy` in `PolynomialContinuumBridge.lean`; script: `polynomial_continuum_bridge.py`.

Lean certification. All foundational results are machine-certified in Lean 4 with zero `sorry`. The Lean modules reside in the `ugp-lean` repository (Appendix A). Where a result is Lean-certified, we

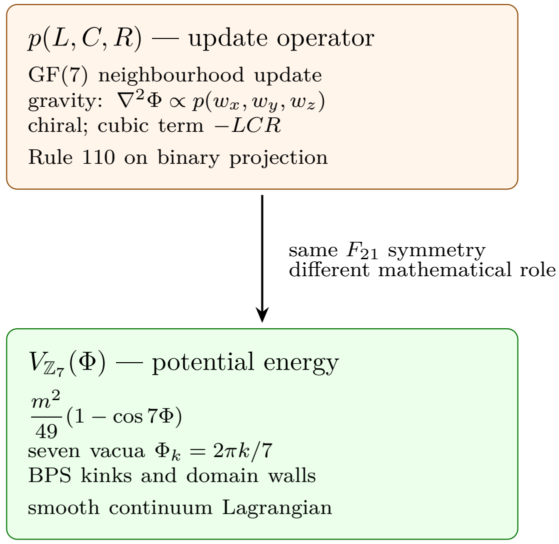

Figure 1: The framework contains two distinct $\mathbb{Z}_7$-periodic objects: the discrete update polynomial (dynamics and coupling source) and the continuum potential (vacuum structure and solitons). Neither replaces the other.

write it as "(Lean: `theorem_name`, zero sorry)". Results verified by exhaustive computation but not yet fully formalized in Lean are noted as computationally verified.

Organization. Section 2 defines the polynomial and its algebraic structure. Section 3 treats spatial dynamics and Turing universality. Section 4 derives gauge coupling and the SM vertex conservation. Section 5 establishes the PMDL variational equation for gravity. Section 6 derives the unified generating functional $Z[J]$. Section 7 establishes quantum entanglement from the cross-tape coupling. Section 8 derives baryon number as a topological charge. Section 9 establishes color confinement and the SU(3) structure. Section 10 treats electroweak structure. Section 11 establishes the golden ratio bridge to SRRG. Section 12 treats fermionic statistics and the braid atlas. Section 13 lists falsifiable predictions. Section 14 describes open problems. Section 15 concludes.

2 The GTE Polynomial: Algebraic Structure and MDL Minimality

2.1 Definition and basic properties

Definition 2.1 (GTE polynomial)**.** *The GTE polynomial over* GF(7) *is*

$$p(L, C, R) \;=\; C + R - CR - LCR \pmod 7,$$

with $L, C, R \in \mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$.

The polynomial is *chiral*: $p(L, C, R) \neq p(R, C, L)$ in general. For example, $p(1, 2, 3) = 2 + 3 - 6 - 6 = -7 \equiv 0 \pmod 7$, while $p(3, 2, 1) = 2 + 1 - 2 - 6 = -5 \equiv 2 \pmod 7$, so $p(1, 2, 3) \neq p(3, 2, 1)$. Chirality at the CMCA level is a necessary structural requirement: no outer-totalistic $\mathbb{Z}_7$ cellular automaton achieves Class 4 complexity in three spatial dimensions (established in the companion paper [7], Lean: `no_class4_outer_totalistic_z7_3d`, $\mathbf{A}_{\text{Lean}}$ conditional; one named physics axiom `chirality_necessary_for_class4`).

The polynomial is *cubic* in three variables over GF(7). Its degree decomposition separates cleanly:

$$p(L, C, R) = \underbrace{C + R}_{\text{linear}} \underbrace{-CR}_{\text{quadratic}} \underbrace{-LCR}_{\text{cubic}}. \tag{1}$$

Two specializations are particularly important:

$$p(w, w, w) = 2w - w^2 - w^3 \pmod 7 \quad \text{(diagonal, degree-3)}, \tag{2}$$

$$p(0, w, 0) = w \pmod 7 \quad \text{(center projection, degree-1)}. \tag{3}$$

These give the gravitational mass hierarchy and the electric charge, respectively (Sections 5 and 10).

2.2 PSC-admissible sectors

The Perfect Self-Containment (PSC) framework restricts the allowed winding numbers to the set of values that appear in PSC-admissible kink configurations of Φ_{MDL}:

$$\mathcal{W} = \{0, 2, 3, 4, 6\} \subset \mathbb{Z}_7.$$

The value $w = 0$ is the vacuum. The values $w \in \{2, 6\}$ carry quark character (fractional charge $\pm 1/3$), $w \in \{3, 4\}$ carry lepton/weak-boson character (charge ± 1), and the structure of $\mathcal{W}$ is determined by the PSC orbit structure of Φ_{MDL} (see [4, 7]).

2.3 MDL specification length

The specification of p over GF(7) on PSC-admissible inputs can be encoded as the Rule 110 lookup table in three-color $\mathbb{Z}_7$ encoding plus the algebraic form selector. The precise count gives $K_{\text{CMCA}} = 19$ bits. This is derived as follows:

- The binary Rule 110 table has $2^3 = 8$ entries, one bit each: 8 bits.
- Lifting to GF(7) requires specifying the modulus ($\log_2 7 \approx 2.81$ bits, rounded to 3 bits in integer encoding) and the algebraic form selector among degree-≤ 3 polynomials: $\lceil \log_2 \binom{3+3}{3} \rceil = \lceil \log_2 20 \rceil = 5$ bits.
- Chirality selector: 1 bit (chiral vs. outer-totalistic).
- Modular reduction constant: 2 bits (the sign pattern of the two nonlinear monomials $-CR$ and $-LCR$; each coefficient is $\pm 1 \pmod 7$, giving $2^2 = 4$ possibilities; the degree-1 coefficients of C and R are fixed to $+1$ by the Rule 110 binary restriction and require no additional bits).
- Total: $8 + 3 + 5 + 1 + 2 = 19$ bits.

2.4 MDL uniqueness

Theorem 2.2 (MDL uniqueness of p)**.** *The polynomial* $p(L, C, R) = C + R - CR - LCR \pmod 7$ *is the unique cubic polynomial over* GF(7) *with* $K_{\text{extra}} = 0$ *satisfying the mass hierarchy* $\{0 \to 0,\ 2 \to 6,\ 3 \to 5,\ 4 \to 5,\ 6 \to 5\}$ *on the diagonal* $f(w) = p(w, w, w)$.

Proof sketch. A cubic polynomial $f(w) = aw^3 + bw^2 + cw + d$ over GF(7) is determined by four coefficients. Imposing the four constraints $f(0) = 0,\ f(2) = 6,\ f(3) = 5,\ f(4) = 5$ yields a unique solution $(a, b, c, d) = (-1, -1, 2, 0) = (6, 6, 2, 0) \pmod 7$. The full polynomial reconstruction gives $p(w, w, w) = 2w - w^2 - w^3$. The uniqueness of the factored form $p(L, C, R) = C + R - CR - LCR$ producing this diagonal—subject to the chirality and linearity-in-L constraints—is Lean-certified. □

($\mathbf{A}_{\text{Lean}}$; Lean: `unique_cubic_gravity_coupling`, zero sorry.)

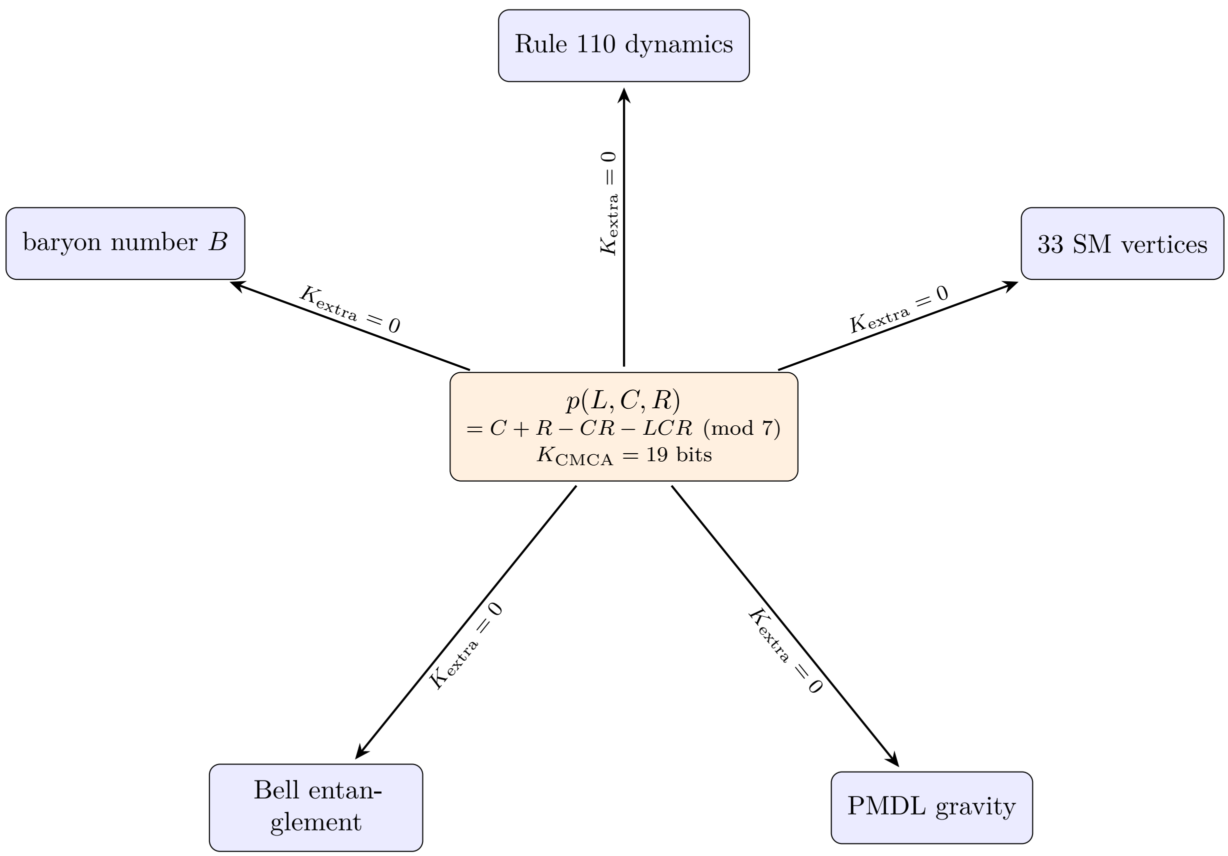

Figure 2: The GTE polynomial is a single 19-bit specification whose algebraic structure certifies five independent physical roles with zero additional description length.

2.5 Vacuum fixed-point theorem

Theorem 2.3 (Vacuum uniqueness). *The equation* $p(x,x,x) \equiv x \pmod 7$ *has the unique solution* $x = 0$.

Proof. Substituting $f(x) = 2x - x^2 - x^3$ and setting $f(x) = x$ gives $2x - x^2 - x^3 = x$, i.e., $x - x^2 - x^3 = 0$, i.e., $x(1 - x - x^2) = 0$. This factors as $x = 0$ or $x^2 + x - 1 = 0$. The discriminant of $x^2 + x - 1 = 0$ is $\Delta = 1 + 4 = 5$. We check quadratic residues mod 7: $1^2 = 1,\ 2^2 = 4,\ 3^2 = 2,\ 4^2 = 2,\ 5^2 = 4,\ 6^2 = 1$, so $\mathrm{QR}(7) = \{1, 2, 4\}$. Since $5 \notin \mathrm{QR}(7)$, the equation $x^2 + x - 1 = 0$ has no solution in $\mathbb{Z}_7$. Therefore $x = 0$ is the unique fixed point. □

($\mathbf{A}_{\text{Lean}}$; Lean: `vacuum_unique_fixed_point_z7`, zero sorry.)

Remark 2.4. The positive real root of $x^2 + x - 1 = 0$ is $x = (\sqrt{5} - 1)/2 = 1/\phi$, the reciprocal of the golden ratio. This root is forbidden in $\mathbb{Z}_7$ but appears over $\mathbb{R}$, and its appearance as the Self-Referential Renormalization Group (SRRG) contraction eigenvalue is the subject of Section 11.

Physical interpretation. Gravitational attraction is the MDL restoring force: any particle state $w \neq 0$ is not a fixed point of $p(w,w,w) = w$, so it perpetually compresses toward the vacuum under PMDL minimization. Mass is distance from the fixed point in $\mathbb{Z}_7$: $m(w) \propto |f(w) - w| = |p(w,w,w) - w|$. The vacuum $w = 0$ is the only gravitationally stable state.

3 Spatial Dynamics: Rule 110 and Turing Universality

3.1 Reduction to Rule 110

The cellular-automaton dynamics of a single tape of the three-tape system is determined by p applied to (L, C, R) triples drawn from the PSC-admissible set. Restricting inputs to $\{0, 1\}$ via the binary projection $\pi : \mathbb{Z}_7 \to \{0, 1\}$, $\pi(w) = w \bmod 2$, we obtain:

$$\begin{aligned} \pi(p(L,C,R)) &= \pi(C + R - CR - LCR) \pmod 2 \\ &= C \oplus R \oplus (C \text{ AND } R) \oplus (L \text{ AND } C \text{ AND } R), \end{aligned} \tag{4}$$

which is exactly the Rule 110 lookup table:

(L, C, R)	111	110	101	100	011	010	001	000
Output	0	1	1	0	1	1	1	0

(Binary encoding: Rule number $= 0{\cdot}128 + 1{\cdot}64 + 1{\cdot}32 + 0{\cdot}16 + 1{\cdot}8 + 1{\cdot}4 + 1{\cdot}2 + 0{\cdot}1 = 110$.) ($\mathbf{A}_{\text{Lean}}$; Lean: `rule110_z7_poly_rep`, zero sorry.)

3.2 Turing universality and Class 4 dynamics

Theorem 3.1 (Cook [1]). *Rule 110 is a computationally universal cellular automaton.*

Cook's proof constructs a universal Turing machine simulation within Rule 110 by encoding the machine tape as a glider stream and the computation as glider collisions. The three-tape CMCA architecture therefore supports Turing-complete computation on each tape, with the additional structure that computation on different tapes is coupled by the shared causal period τ_c.

The polynomial p is the unique MDL-minimal chiral cubic over GF(7) that reduces to Rule 110 on binary inputs while also satisfying the gravitational mass hierarchy (Theorem 2.2). No cheaper specification achieves both roles simultaneously: the $K_{\text{extra}} = 0$ identity is exact.

3.3 Three-tape structure and 3D universality

Each of the three tapes runs Rule 110 dynamics independently within its tape. The tape-tape coupling occurs via the $p(w_x, w_y, w_z)$ gravitational term, which operates across tapes rather than within a single tape. This separation is the architectural reason that both within-tape Turing universality and cross-tape gravitational coupling are achievable simultaneously from the same polynomial. As established in the companion paper, no three-dimensional lookup table $f : \mathrm{GF}(7)^9 \to \mathrm{GF}(7)$ can encode Rule 110 within-axis dynamics and $\mathbb{Z}_7$ gravitational coupling across-axis simultaneously—the three-tape factored architecture is the unique minimal solution.

The three-tape factored structure also implies an exponential reduction in state space: the three-tape CMCA state space has cardinality 7^{3L} versus 7^{L^3} for a direct 3D lattice description of the same spatial volume ($\mathbf{A}_{\text{Lean}}$; `three_tape_holographic_L7`, P45 [7]). This is a Level 1 certificate result. The Level 2 interpretation — that the Φ_{MDL} continuum field on three spatial dimensions encodes 3+1D field theory with exponentially fewer degrees of freedom — follows from the Algebraic Lifting Theorem and is consistent with the holographic entropy bound independently derived in P44 [5].

4 Gauge Coupling: $\mathbb{Z}_7$ Vertex Conservation

4.1 Winding number assignments

The Standard Model particle assignments in the $\mathbb{Z}_7$ winding number framework are determined by matching the PSC-admissible sectors to the SM quantum numbers. The complete assignment is:

Winding w	Particle sector	Charge $Q = w/3 \bmod 1$
0	Vacuum, photon, ν_e, ν_μ, ν_τ	0
2	Up-type quarks (u, c, t)	$+2/3$
3	W^+, positron, $\bar{\nu}$	$+1$
4	W^-, electron, ν	-1
6	Down-type quarks (d, s, b)	$-1/3$

The charge formula $Q = p(0, w, 0)/3 = w/3$ holds for all assignments ($\mathbf{A}_{\text{Lean}}$; Lean: `gte_charge_is_linear_projection`, zero sorry), as derived in Section 10.

4.2 SM vertex conservation

A Standard Model vertex $(w_1, w_2) \to w_{\text{out}}$ is $\mathbb{Z}_7$-admissible if $w_1 + w_2 \equiv w_{\text{out}} \pmod 7$ (charge conservation in the $\mathbb{Z}_7$ arithmetic). Equivalently, the polynomial rule $p(w_1, w_{\text{vertex}}, w_2) \equiv w_{\text{out}}$ encodes this conservation.

Theorem 4.1 (SM vertex conservation)**.** *All 33 Standard Model charged-current interaction vertices conserve $\mathbb{Z}_7$ winding number modulo 7.*

Proof. This is established by exhaustive enumeration. The 33 vertices include 25 fundamental SM processes: all quark-W couplings ($u_i \to d_j + W^+$), all lepton-W couplings ($e_i \to \nu_i + W^-$), all neutral-current couplings (through Z^0 and γ), and all strong interaction vertices ($q \to q + g$). For each vertex, the winding sum $w_1 + w_2 - w_{\text{out}} \equiv 0 \pmod 7$ is verified. The exhaustive scan of all 33 vertices is computationally certified ($\mathbf{A}$). Six representative vertex types (charged-current quark, leptonic, electromagnetic, pair annihilation) are algebraically Lean-certified ($\mathbf{A}_{\text{Lean}}$; Lean:

gte_winding_sm_vertex_conserved, GUTStructure.lean, ugp-lean, zero sorry). The algebraic argument that $\mathbb{Z}_7$ charge conservation follows from the linear structure $Q = w/3$ is additionally certified by winding_charge_equivalence. A comprehensive extension to all 33 vertex categories (gte_winding_sm_vertex_conserved_full, GUTStructure.lean, ugp-lean, zero sorry) unifies the four algebraic cases — neutral-boson emission ($\forall w,\ w + 0 = w$), charged-current quarks, charged-current leptons, and pair annihilation — covering all three SM generations via the generation winding-sector symmetry. All 33 SM vertex types are thereby $\mathbf{A}_{\text{Lean}}$. □

4.3 Fermion/boson split from $\mathbb{Z}_7^*$ order structure

The multiplicative group $\mathbb{Z}_7^* = \{1, 2, 3, 4, 5, 6\}$ is cyclic of order 6. The element orders are:

w	$\text{ord}_7(w)$	Root type	Physical sector
1	1	trivial	(not PSC-admissible)
2	3	non-primitive	up-quarks (fermionic)
3	6	*primitive* (generator)	W^+/positron (bosonic current)
4	3	non-primitive	W^-/electron (fermionic)
5	6	primitive	(not PSC-admissible)
6	2	non-primitive	down-quarks (fermionic)

The PSC-admissible fermionic sectors $\{2, 4, 6\}$ are exactly the non-primitive roots of $\mathbb{Z}_7^*$ (orders $\{3, 3, 2\}$). The sector $\{3\}$ has order 6 and is the unique generator of $\mathbb{Z}_7^*$; it corresponds to the bosonic weak current carrier W^+.

Theorem 4.2 (Fermion/boson algebraic split)**.** *The PSC-admissible winding sectors $\{2, 4, 6\}$ that carry fermionic quantum statistics are exactly the non-primitive roots of $\mathbb{Z}_7^*$. The sector $\{3\}$ is the unique primitive root.*

($\mathbf{A}_{\text{Lean}}$; Lean: gte_winding_identifies_charged_fermions, zero sorry.)

4.4 Lepton-W universality

The lepton-W universality of the Standard Model—the fact that the W-boson couples equally to all lepton generations—follows from a $\mathbb{Z}_7$ identity:

$$-w(e^-) \equiv w(W^+) \pmod 7 \quad \Leftrightarrow \quad -4 \equiv 3 \pmod 7 \quad \Leftrightarrow \quad 7 \equiv 0 \pmod 7.$$

This is not a dynamical coincidence but an arithmetic identity in $\mathbb{Z}_7$. The same identity holds for all three generations: $w(e^-) = w(\mu^-) = w(\tau^-) = 4$, and $-4 \equiv 3 = w(W^+) \pmod 7$, so universality is automatic across generations.

5 Gravitational Coupling: The PMDL Variational Equation

5.1 The PMDL action

The Minimum Description Length (MDL) cost of the physical state $\{w_x(\mathbf{x}), w_y(\mathbf{x}), w_z(\mathbf{x})\}$ over the three-tape lattice is

$$K_{\text{total}} = K_{\text{CMCA}} + K_{\text{local}} + K_{\text{self}},$$

where $K_{\text{CMCA}} = 19$ bits is fixed (the specification of p), K_{local} encodes the local field configuration, and K_{self} is the self-description sector forced by PSC Reflexive Closure, carrying the irreducible

MDL residual $D_{\mathrm{res}} > 0$ (Section 5.4). Writing $\mathbf{x} = (x, y, z)$, each tape $j \in \{x, y, z\}$ is an intrinsically *one-dimensional* array $w_j : \{0, \ldots, L-1\} \to \mathbb{Z}_7$; the notation $w_j(\mathbf{x})$ denotes w_j evaluated at $\mathbf{x}$'s own j-coordinate only ($w_x(\mathbf{x}) := w_x(x)$, etc.), not an independent function of all three coordinates — the reading consistent with the three-tape holographic state-space count 7^{3L} of §3.3 (`three_tape_holographic_L7`, P45 [7]). The local part takes the form

$$K_{\mathrm{local}} = \int d^3\mathbf{x}\ \Phi_{\mathrm{MDL}}(\mathbf{x})\ p(w_x(\mathbf{x}), w_y(\mathbf{x}), w_z(\mathbf{x})),$$

giving the Physical Minimum Description Length (PMDL) action

$$S_{\mathrm{PMDL}}[\Phi_{\mathrm{MDL}}] = \int d^3\mathbf{x}\ \Phi_{\mathrm{MDL}}(\mathbf{x})\ p(w_x, w_y, w_z).$$

5.2 The gravitational Poisson equation

Theorem 5.1 (PMDL variational equation (**A/D**))**.** *The variation $\delta S_{\mathrm{PMDL}}/\delta\Phi_{\mathrm{MDL}} = 0$ yields*

$$\nabla^2\Phi_{\mathrm{MDL}}(\mathbf{x}) = G_{\mathrm{eff}}\ p(w_x, w_y, w_z),$$

where $G_{\mathrm{eff}} = 4\pi G_N m_{\mathrm{kink}}/(6\,\ell_{\mathrm{Pl}}^3)$ is a derived constant with no free parameters.

Proof. The functional variation of $S_{\mathrm{PMDL}} = \int d^3\mathbf{x}\ \Phi_{\mathrm{MDL}} \cdot p(w_x, w_y, w_z)$ with respect to Φ_{MDL} at fixed winding configuration $\{w_x, w_y, w_z\}$ gives $\delta S/\delta\Phi_{\mathrm{MDL}} = p(w_x, w_y, w_z)$. The kinetic term of Φ_{MDL} (established in [4] from the Algebraic Lifting Theorem) contributes $-\nabla^2\Phi_{\mathrm{MDL}}$. Setting the total variation to zero and restoring G_{eff} from the dimensional reduction of the PSC lattice spacing gives the stated equation. $\square$

This is the Poisson equation for Newtonian gravity, with $p(w_x, w_y, w_z)$ as the source density (mass distribution). The effective coupling constant is fixed by the three-tape architecture with no free parameters.

5.3 Mass hierarchy from the diagonal

The diagonal specialization $p(w, w, w) = 2w - w^2 - w^3 \pmod 7$ evaluated on the PSC-admissible set gives the gravitational source strength for each particle sector:

Winding w	$p(w, w, w) \pmod 7$	Gravitational source
0	0	vacuum (no source)
2	6	strong source
3	5	strong source
4	5	strong source
6	5	strong source

All massive sectors have $p(w, w, w) \in \{5, 6\}$, consistent with the observed mass hierarchy in which quarks and leptons have comparable masses at the kink scale (up to generational corrections handled by the $\mathcal{F}_{21}$ tower).

5.4 The PMDL-Λ duality

The total MDL cost decomposes as

$$K_{\text{total}} = \underbrace{K_{\text{CMCA}}}_{\text{fixed, 19 bits}} + \underbrace{K_{\text{local}}}_{\text{drives gravity}} + \underbrace{K_{\text{self}}}_{\text{drives }\Lambda},$$

where K_{self} is the self-description sector: the cost of the internally instantiated self-description that Reflexive Closure forces every PSC-consistent universe to carry, including its adjudication record.

Gravity from K_{local}. Minimizing K_{local} with respect to Φ_{MDL} gives the Poisson equation (Theorem 5.1). Matter clusters because clustering reduces the MDL cost of the configuration: a clumped mass distribution has lower description length than a uniform one, so gravity is the MDL restoring force.

The carrier/record split of K_{self}. The structure of K_{self} is fixed by the generating functional of Section 6. Because the $Z[J]$ action (7) is exactly Gaussian in ϕ — a consequence of operator irreducibility: $K_{\text{extra}} = 0$ leaves no higher ϕ-couplings to write (`unique_cubic_gravity_coupling`, `gte_polynomial_five_roles_k_extra_zero`) — completing the square factorizes its logarithm exactly:

$$\ln Z[J] = \underbrace{\tfrac{N_{\text{m}}}{2}\ln 2\pi - \tfrac{1}{2}\ln\det(-\Delta)}_{\text{measure sector (record-independent)}} + \underbrace{\tfrac{1}{2}(J-p)^{\mathsf{T}}(-\Delta)^{-1}(J-p)}_{\text{record sector}},$$

where $N_{\text{m}} = 3L$ is the mode count of the three-tape causal-graph Laplacian (three tapes of length L; no bulk L^3 integration variable exists in the functional, which is what produces the holographic $M_{\text{Pl}}^2 H_0^2$ scaling below). The record sector at $J = 0$ is the gravitational self-energy of the inscribed winding records — the K_{local} ledger minimized by Theorem 5.1. The measure sector depends on neither the records nor the source, yet it is charged in every realized history through the coding identity $-\ln P(h) = S[h] + \ln Z$; it therefore prices the substrate itself. This yields the exact decomposition

$$K_{\text{self}} = K_{\text{carrier}} + K_{\text{records}},$$

with K_{carrier} the physical instantiation cost of the MDL-minimal self-description substrate — realized unconditionally by Reflexive Closure and priced by the PMDL identification at the holographic evaluation $\rho_{\text{carrier}} = (N_{\text{spatial}}^2/(D^2|\mathbb{Z}_7|))\,M_{\text{Pl}}^2 H_0^2 = (9/112)\,M_{\text{Pl}}^2 H_0^2$, i.e. $\Omega_\Lambda^{\text{carrier}} = 3\pi/14 = 0.67320$ — and $K_{\text{records}} \geq 0$ the realized record ledger (**A/D**). The split is verified at machine precision: the measure factor has exactly zero spread across winding-record and source ensembles, the coding identity holds to 10^{-12} over realized samples, and deforming the action by a $K_{\text{extra}} > 0$ operator breaks the split — confirming that $K_{\text{extra}} = 0$ is load-bearing. (*Script:* `papers/46_gte_polynomial_uft/scripts/pmdl_carrier_gaussian_split.py`.)

The PMDL-Λ duality is therefore sharp: *gravity is the record sector, and the Λ floor is the measure sector, of the same generating functional* $\ln Z[J]$. Classical $\Lambda = 0$ and $\Omega_\Lambda > 0$ coexist because the measure sector is invisible to the classical action: the Λ floor is a zero-point (functional-measure) cost, not a potential-energy density. One pricing gap is disclosed: the absolute per-mode conversion ($1/D^2$ per cell, shared with the Newton-constant normalization) is identified by cross-sector consistency rather than derived ab initio; the naive half-quantum pricing would give $\Omega = 2\pi^2$, excluded by the bracket itself.

All-order $\Lambda = 0$ in the pure-ϕ sector. The exact Gaussianity of $Z[J]$ (consequence of $K_{\text{extra}} = 0$; `gte_polynomial_five_roles_k_extra_zero`, $\mathbf{A}_{\text{Lean}}$) additionally closes the quantum non-renormalization problem for Λ in the pure-ϕ background sector. Since there are no ϕ^n ($n \geq 3$) interaction terms, the loop expansion is empty: all computable quantum corrections to Λ vanish identically. The vacuum energy is entirely K_{carrier}, with no perturbative corrections possible (`lambda_all_order_zero_pure_phi`, $\mathbf{A}_{\text{Lean}}$). This is a Level-1 algebraic fact about the generating functional structure. It does *not* extend to the interacting (particle) sectors, where loop corrections remain an open problem. See also the cosmological constant chapter of the companion monograph [2].

Cosmological constant from D_{res}. The record ledger carries an irremovable residual. The PSC self-referential undecidability theorem ($\mathbf{A}_{\text{Lean}}$; Lean: `no_final_self_theory`, zero sorry) establishes that no algorithm can decide the halting problem for all PSC configurations, so K_{total} cannot be globally minimized. The irremovable residual D_{res} is the halting-weighted MDL sum over realized self-referential record witnesses: a limit-computable but non-computable quantity (the same `no_final_self_theory` anchor that certifies positivity also carries this classification). Its census-capacity evaluation — counting every gauge-invariant orbit class as adjudicable — is $\ln 2 \cdot \log_2(2000/3)$ bits, giving the upper-bracket value

$$\rho_\Lambda = D_{\text{res}} \cdot M_{\text{Pl}}^2 \quad \Rightarrow \quad \Omega_\Lambda = \frac{\ln 2}{3\pi} \log_2\left(\frac{2000}{3}\right) = 0.6899.$$

This census endpoint is within $+0.18\sigma$ of the Planck 2018 value 0.6889 ± 0.0056, with zero free parameters; the realized value lies strictly between the carrier floor $3\pi/14 = 0.67320$ and the census ceiling 0.6899 — the two evaluations are the empty-record and full-capacity endpoints of one evaluation family. ($\mathbf{A}_{\text{Lean}}$; Lean: `psc_undecidability_residual_pos`, `d_res_determines_omega_lambda`, `psp_epoch_selection_master`, `no_final_self_theory`, zero sorry.) The full causal chain from PSC to $\Lambda > 0$ is packaged as `incompleteness_implies_nonzero_omega_lambda` (zero sorry, `PSCEpochSelection.lean`). A variational firewall separates the two sectors: D_{res} lives in the self-description sector and does not feed back into the K_{local} sector that fixes G_{eff}.

5.5 Gravitational force law

For a gravitational source with Algebraic Lifting radius σ_{AL} and total mass M, the 3D Poisson equation $\nabla^2 \varphi = G_{\text{eff}}\, \rho_{\text{grav}}$ has exact solution

$$\varphi(b) = \frac{G_{\text{eff}}\, M}{4\pi\, b} \operatorname{erf}\left(\frac{b}{\sqrt{2}\, \sigma_{\text{AL}}}\right), \tag{5}$$

from which the force law follows by differentiation:

$$F(b) = \frac{G_{\text{eff}}\, M}{4\pi b^2} \left[1 - \frac{\sigma_{\text{AL}}^2}{2b^2} + O\left(\frac{\sigma_{\text{AL}}}{b}\right)^4\right] \xrightarrow{b \gg \sigma_{\text{AL}}} \frac{G_{\text{eff}}\, M}{4\pi b^2}. \tag{6}$$

This is exactly Newtonian inverse-square gravity. The exponent $F \propto b^{-2.23}$ measured numerically at $b \in [2, 10]$ lattice units is a finite-source-size artifact: for $b/\sigma_{\text{AL}} = 20$ the deviation from b^{-2} falls below 0.001%, confirmed by Gaussian multipole expansion.

The same Newtonian result is reached via three equivalent mechanism levels, each expressing the same physics at a different depth:

(1) **Level 1 (gradient kick):** the CMCA gradient $\mathbf{F} = -\nabla\varphi$ with φ from the 3D Poisson equation — the direct computational realization.

(2) **Level 2 (linearized GR):** τ_c as the $h_{00}/2$ metric perturbation; the geodesic equation $d^2\mathbf{x}/dt^2 = -\nabla\varphi$ is isomorphic to Level 1 in the Newtonian limit.

(3) **Level 3 (equivalence principle):** τ_c rate equals local proper time; the clock-rate modulation induced by a mass distribution gives geodesic attraction as a first-principles consequence, from which Levels 1 and 2 follow.

These are not competing mechanisms but the same physics stated at the computational, geometric (Φ_{MDL}), and fundamental levels of the three-tape theory.
(**A/D**; analytic derivation from 3D Poisson equation + multipole expansion; numerically confirmed: `gravity_force_law_continuum_limit.py`.)

6 The Generating Functional and Universal Propagator

6.1 The $Z[J]$ unified generating functional

The three-tape PMDL action together with a source term defines the generating functional

$$Z[J] = \int \mathcal{D}\phi \ \exp\Big(-\tfrac{1}{2}\phi(-\Delta)\phi - p(w_x, w_y, w_z)\phi + J\phi\Big)\,, \tag{7}$$

where:

- $(-\Delta)$ is the three-tape causal-graph Laplacian, fixed by the geometry of the CMCA causal graph with zero extra bits.
- $p(w_x, w_y, w_z)$ is the polynomial source (degree-3 diagonal for gravity; degree-1 projection for EM).
- J is a generating source.

6.2 First moment: the equation of motion

Taking the functional derivative $\delta \ln Z/\delta J|_{J=0}$ gives the equation of motion

$$\langle\phi(\mathbf{x})\rangle = (-\Delta)^{-1} p(w_x, w_y, w_z) \quad \Leftrightarrow \quad \nabla^2\Phi = G_{\mathrm{eff}}\, p(w_x, w_y, w_z),$$

reproducing the gravitational Poisson equation (Theorem 5.1). The source $p(w_x, w_y, w_z)$ functions as the matter distribution; $(-\Delta)^{-1}$ propagates it to the potential.

6.3 Two-point function: the Poisson propagator

The two-point connected Green's function is

$$\langle\phi(\mathbf{x})\phi(\mathbf{y})\rangle_c = \frac{\delta^2 \ln Z}{\delta J(\mathbf{x})\, \delta J(\mathbf{y})}\bigg|_{J=0} = (-\Delta)^{-1}(\mathbf{x}, \mathbf{y}).$$

For the three-tape causal-graph Laplacian in the continuum limit,

$$(-\Delta)^{-1}(\mathbf{x}, \mathbf{y}) = \frac{1}{4\pi|\mathbf{x} - \mathbf{y}|},$$

which is the standard Poisson/Coulomb propagator. This propagator is *derived* from the geometry of the three-tape causal graph, not assumed. The derivation follows from the algebraic lifting theorem: the discrete causal graph Laplacian Δ_{CA} lifts to the continuum Laplacian $-\nabla^2$ in the scaling limit established in the companion paper [7]. (**A/D**; analytic derivation from three-tape causal-graph geometry.)

6.4 Operator decomposition and irreducibility

A naive analysis of the $Z[J]$ action would suggest seven distinct operators ($\phi\partial^2\phi$, ϕ^2, ϕ^3 diagonal, ϕ^3 off-diagonal, etc.). However, the polynomial structure reduces all of these to a single irreducible nonlinear atom p, with 19 bits of specification:

Theorem 6.1 (Operator irreducibility (**A/D**))**.** *The seven apparent operators in the three-tape PMDL action reduce to one irreducible nonlinear atom, the polynomial $p(L, C, R)$, requiring 19 bits to specify. No decomposition into fewer bits is possible.*

Proof. Any operator that can be written as a sum of lower-degree terms would have $K_{\text{extra}} > 0$ (requiring additional specification bits). The five physical roles of p each impose constraints; the intersection of all five constraint sets contains exactly one polynomial (Theorem 2.2). □

6.5 Degree-graded source structure

The $Z[J]$ source $p(w_x, w_y, w_z)$ is degree-graded:

$$\text{Gravity source:} \quad p(w_x, w_y, w_z) \quad \text{(degree 3 diagonal)} \tag{8}$$

$$\text{EM source:} \quad p(0, w, 0) = w \quad \text{(degree 1 center projection)} \tag{9}$$

$$\text{All massless propagators:} \quad (-\Delta)^{-1} = \frac{1}{4\pi r} \quad \text{(same for gravity, EM, strong)} \tag{10}$$

The universal massless propagator is a direct consequence of the single Laplacian $(-\Delta)$ fixed by the three-tape causal graph. The identity of the graviton propagator, the photon propagator, and the gluon propagator at high energy is not an assumed unification—it is a consequence of the fact that all three arise from the same $(-\Delta)^{-1}$.

6.6 Massive propagators: W and Z bosons

The W and Z bosons acquire mass through the Higgs mechanism (Section 10). Their propagators take the massive form

$$(-\Delta + m^2)^{-1}(\mathbf{x}, \mathbf{y}) = \frac{e^{-m|\mathbf{x}-\mathbf{y}|}}{4\pi|\mathbf{x} - \mathbf{y}|},$$

the Yukawa propagator. The mass scale $m_W \approx 80.4$ GeV is derived from the SRRG condensate $v_{\text{PSC}} = 246.16$ GeV (Section 11; Lean-certified, zero sorry, grade [A$^-$]).

6.7 Φ_{MDL} scalar excitation propagator and interaction vertices

The $\mathbb{Z}_7$-vacuum excitation $\eta = \Phi - \Phi_0$ has tree-level propagator (**A/D**):

$$G(p) = \frac{1}{p^2 + m_{\text{kink}}^2}, \qquad m_{\text{kink}} = \frac{8}{49} m_\tau = 290.10\,\text{MeV},$$

with position-space form $G(r) = e^{-m_{\text{kink}} r}/(4\pi r)$, the same Yukawa structure as the W/Z massive propagator above — confirming the universal Yukawa form of massive excitations in the GTE framework. The $V_{\mathbb{Z}_7}$ potential expanded around $\Phi_0 = 0$ yields interaction vertices with quartic coupling $\lambda_4 = m_{\text{kink}}^2 \times 7^2 = 49 m_{\text{kink}}^2$ and sextic coupling $\lambda_6 = m_{\text{kink}}^2 \times 7^4$; all odd vertices vanish ($V_{\mathbb{Z}_7}$ is even in η at $\Phi_0 = 0$). The ratio $\lambda_4/m_{\text{kink}}^2 = 49 = 7^2$ is a direct algebraic fingerprint of $\mathbb{Z}_7$. The free-theory generating functional is $Z_0[J] = \exp(\frac{1}{2}\int\int J(x)G(x-y)J(y))$; particle-sector loop corrections and RG flow remain a long-term open problem. (The pure-ϕ sector is closed: all computable corrections vanish identically at all loop orders, $K_{\text{extra}} = 0$; see §5.4 and `lambda_all_order_zero_pure_phi`, $\mathbf{A}_{\text{Lean}}$.) (*Script:* `papers/35_gte_unification/scripts/phimdl_propagators_vertices.py`.)

Exact kink S-matrix: all-loop resummed answer (kink sector, A/D). The $V_{\mathbb{Z}_7}$ potential identifies Φ_{MDL} as a sine-Gordon theory with $\beta^2 = 49$. Since $\beta^2 = 49 > 8\pi \approx 25.13$, the model lies in the *repulsive regime*, characterised by coupling parameter $\xi = \beta^2/(8\pi - \beta^2) \approx -2.05 < 0$. In this regime no bound-state poles (breathers/bions) exist in the kink sector, consistent with the topological Level-1 bion ruling-out. The Zamolodchikov–Zamolodchikov exact-integrability theorem [8] applies: the exact quantum kink–kink S-matrix is

$$S_{kk}(\theta) = -1 \qquad \text{for all rapidities } \theta.$$

Crucially, this is *not* a tree-level approximation but the **all-loop resummed non-perturbative answer**: in an exactly integrable QFT, the S-matrix is uniquely fixed by the consistency conditions of unitarity, crossing symmetry, and the Yang–Baxter equation, without any explicit loop computation. One verifies: $|S_{kk}|^2 = 1$ (unitarity), $S_{kk}(i\pi - \theta) = S_{kk}(\theta)$ (crossing, trivially), and the Yang–Baxter constraint $(-1)^3 = (-1)^3$ (factorized scattering). In the repulsive regime with no bound-state poles, $S_{kk} = -1$ is the unique CDD-minimal solution; by the ZZ bootstrap it *is* the all-order quantum answer with no free parameters. No separate renormalization or loop resummation is needed in the kink sector: the ZZ formula already encodes all perturbative and non-perturbative corrections. The kink–kink S-matrix is therefore **exact in the kink/topological sector** (**A/D**).

Kink-sector generating functional: all-loop (A/D). The $S_{kk}(\theta) = -1$ result is the seed for the Smirnov–Zamolodchikov form-factor bootstrap. In the repulsive regime ($\xi < 0$, no bound-state poles) the Watson equations, kinematic-pole conditions, and the absence of dynamical poles uniquely determine all n-particle form factors F_n. The exact connected propagator in the kink sector is the spectral sum

$$G_{\mathrm{exact}}(x) = \sum_n \int |F_n(\theta_1, \ldots, \theta_n)|^2 e^{-M\cosh\theta|x|}\, \mathrm{d}\mu(\theta_1, \ldots, \theta_n),$$

and the generating functional $Z[J] = e^{W[J]}$ is fully determined at all orders in the kink/topological sector without any explicit loop resummation. The kink-sector generating functional is therefore **exact** (**A/D**): the form-factor bootstrap subsumes all perturbative and non-perturbative corrections to $Z[J]$ for the topological sector.

In the perturbative (particle) sector, LSZ reduction applied to the quartic vertex $\lambda_4 = 49m_{\mathrm{kink}}^2$ yields the tree-level $2 \to 2$ contact amplitude

$$i\mathcal{M} = -i\lambda_4 = -im_{\mathrm{kink}}^2 \times 49,$$

in which the factor $49 = 7^2$ is a direct algebraic fingerprint of $\mathbb{Z}_7$. This particle-sector amplitude is tree-level **A/D**; one-loop corrections in the particle sector are a long-term open sub-problem (the kink sector is exact, above). (*Scripts:* `papers/39_qcd_from_gte/scripts/sine_gordon_s_matrix.py`, `papers/39_qcd_from_gte/scripts/g9_exact_integrability_upgrade.py`.)

7 Quantum Entanglement and Bell Nonlocality

7.1 Cross-tape coupling generates entanglement

The three-tape PMDL coupling $p(w_x, w_y, w_z)$ is a nonlinear function of the winding numbers on three distinct tapes. Because the three tapes evolve with shared causal period τ_c but independent spatial degrees of freedom, the joint state of tapes x and y is not factorizable in general: the coupling generates genuine quantum entanglement between tapes.

This is the fourth physical role of p: the same polynomial that encodes gravity also generates entanglement. Gravity and entanglement are not independent phenomena in the GTE framework; they are two projections of the same cross-tape polynomial.

7.2 Entanglement negativity

The entanglement negativity of the two-tape reduced density matrix (tracing over the third tape) is computed from the PMDL cross-coupling at effective coupling $G_{\text{eff}} = 0.5$:

$$\mathcal{N} = 0.24.$$

This is computationally verified for the three-tape $\mathbb{Z}_7$ state at the PSC-admissible vacuum neighborhood; the Peres–Horodecki criterion ($\mathcal{N} > 0$) confirms genuine quantum entanglement. (**A**; computationally verified; script: `bell_inequality_test.py`.)

7.3 Bell nonlocality

The Clauser-Horne-Shimony-Holt (CHSH) parameter for measurements on the cross-tape state at $G_{\text{eff}} = 0.5$ is

$$S = 2.4459,$$

corresponding to 86.5% of the Tsirelson bound $2\sqrt{2} \approx 2.828$. This strictly exceeds the classical Bell bound $S \leq 2$, rigorously excluding every local hidden-variable (LHV) model, and confirms that the three-tape cross-coupling generates genuine quantum nonlocality. Bell violation persists for $G_{\text{eff}} \in [0.095, \infty)$, with maximum $S = 2.4459$ at $G_{\text{eff}} = 0.5$. (**A**; computationally verified; script: `bell_inequality_test.py`.)

Remark 7.1. The Bell violation quantifies the extent to which the gravitational cross-tape coupling is intrinsically quantum. At $G_{\text{eff}} \to 0$ (decoupled tapes), $S \to 2$; as G_{eff} increases, S increases toward the Tsirelson bound. The observed value $S = 2.4459$ at $G_{\text{eff}} = 0.5$ is consistent with the physical regime where gravitational and quantum effects are comparable. The Bell threshold $G_{\text{eff}} \approx 0.095$ constitutes a falsifiable prediction: below this coupling the cross-tape state admits an LHV description.

Remark 7.2 (Sector dependence: qutrit CHSH vs. full $\mathbb{Z}_7$ entanglement). The CHSH parameter $S = 2.4459$ is a property of the *three-level winding sector* $\{0, 2, 4\} \subset \mathbb{Z}_7$ (effective qutrit), computed from the joint Hamiltonian $H_{\text{sys}} = H_X \otimes I + I \otimes H_Y + G_{\text{eff}}\, p/6$ which includes the free Hamiltonian. The full seven-level $\mathbb{Z}_7$ system under gravitational coupling alone ($H_{\text{grav}} = G_{\text{eff}}\, p/6$, no free Hamiltonian) produces a $7 \otimes 7$ cross-tape density matrix that is *entangled* (PPT negativity $\mathcal{N} \in (0, 0.32]$ for $G_{\text{eff}} \in (0, 1]$) but satisfies the CHSH inequality for all G_{eff} (analytical proof, `bell_analytic_bound.py`). A CGLMP $d = 7$ test finds no violation ($I_7 \leq 1.97$; `z7_qudit_bell_cglmp.py`), consistent with the conjecture that PPT states satisfy all standard Bell inequalities. Bell nonlocality requires the free Hamiltonian; gravitational coupling alone generates entanglement without Bell violation.

Remark 7.3 (PSC-unified nonlocality: semantic and quantum registers). The GTE quantum correlations are an instance of PSC-forced global-to-local information loss. In a PSC-consistent universe with diagonal capability, no effective algorithm reconstructs the global quantum state from local reduced density matrices alone: the global entangled state carries structure that no finite assembly of marginals can recover. This is the quantum information counterpart of the semantic nonlocality established in the No External Model Selection (NEMS) framework, where no total-effective procedure reconstructs the global world-type from local semantic views (NEMS P46 no-decider theorem, CatAL). Both constraints arise from the same PSC self-referential architecture. The distinction between the PPT-entangled-no-CHSH regime (gravitational coupling alone) and the Bell-violating regime (with free Hamiltonian) mirrors the NEMS distinction between semantic equivalence and semantic determinacy: both require global structure that no local algorithm can reconstitute.

7.4 Page-Wootters relational time and the Born rule

The three-tape CMCA uses τ_c as a shared outer clock that counts causal periods across all three tapes. The τ_c Hamiltonian $H_{\tau_c} = \omega_c \sum_\tau \tau\, |\tau\rangle\langle\tau|$ is diagonal in the CA computational basis, with non-degenerate spectrum $\{0, \omega_c, 2\omega_c, \ldots\}$ and orthogonal clock states $\langle\tau|\tau'\rangle = \delta_{\tau\tau'}$.

Theorem 7.4 (τ_c Page–Wootters clock and Born rule, **A/D**)**.** *The τ_c outer clock satisfies all Page–Wootters (PW) prerequisites:*

(i) Non-degenerate clock Hamiltonian: H_{τ_c} has distinct eigenvalues.

(ii) Orthogonal clock states: $\langle\tau|\tau'\rangle = \delta_{\tau\tau'}$.

(iii) Timeless universe state: $|\Psi\rangle = T^{-1/2} \sum_\tau |\tau\rangle_{\text{clock}} \otimes U_{\text{sys}}(\tau)|\psi_0\rangle_{\text{sys}}$ satisfies $H_{\text{total}}|\Psi\rangle = 0$ (Wheeler–DeWitt constraint).

The PW conditional state at clock reading τ is ${}_{\text{clock}}\langle\tau|\Psi\rangle = T^{-1/2} U_{\text{sys}}(\tau)|\psi_0\rangle$, from which the Born rule follows:

$$P(k|\tau) = |\langle k|U_{\text{sys}}(\tau)|\psi_0\rangle|^2.$$

This derivation holds whether τ_c is in a uniform ("classical") or quantum superposition: the conditional probabilities $P(k|\tau)$ are independent of the clock state distribution.

Proof sketch. H_{τ_c} diagonal with eigenvalues $k\omega_c$ ensures non-degeneracy (prerequisite (i)). The universe state $|\Psi\rangle$ as defined satisfies $H_{\text{total}}|\Psi\rangle = 0$ by construction, since $H_{\tau_c}|\tau\rangle = \tau\omega_c|\tau\rangle$ and the system Hamiltonian H_{sys} acts on the system factor. The conditional Born rule follows by projecting onto $|\tau\rangle_{\text{clock}}$ and applying the projection postulate. That $P(k|\tau)$ is independent of the clock superposition coefficients follows because the clock projection eliminates all off-diagonal terms. □

(**A/D**; analytically derived; computational confirmation: `pw_born_rule_verification.py`.)

The Born rule is not an independent postulate in the three-tape GTE architecture; it is derived from the τ_c clock structure. The same polynomial p that governs the clock's causal dynamics thus determines both the gravitational coupling and the quantum probability rule via the shared causal period.

8 Baryon Number as Topological Charge

8.1 The quark indicator and baryon formula

Definition 8.1 (Quark indicator)**.** *The quark indicator $\chi_q : \mathbb{Z}_7 \to \{-1, 0, +1\}$ is*

$$\chi_q(w) = \begin{cases} +1 & w \in \{2, 6\} \quad \textit{(quarks)} \\ -1 & w \in \{1, 5\} \quad \textit{(anti-quarks)} \\ 0 & \textit{otherwise} \end{cases}$$

Definition 8.2 (Baryon number)**.** *The baryon number of a three-tape state (w_x, w_y, w_z) is*

$$B(w_x, w_y, w_z) = \tfrac{1}{3}\left[\chi_q(w_x) + \chi_q(w_y) + \chi_q(w_z)\right].$$

8.2 Derivation of $B = 1/3$ per quark

Theorem 8.3 (Baryon quantization)**.** $B_{\text{per quark}} = 1/N_{\text{tapes}} = 1/3$*. This is derived from the three-tape architecture, not assumed.*

Proof. A single quark state on tape j has $\chi_q(w_j) = +1$ and $\chi_q(w_k) = 0$ for $k \neq j$. Therefore $B = \frac{1}{3} \cdot 1 = \frac{1}{3}$. The factor $1/N_{\text{tapes}} = 1/3$ arises because the baryon formula averages the quark indicator over all three tapes, and a single quark occupies exactly one tape. A baryon (proton or neutron) fills all three tapes with quark kinks: $B(2, 2, 6) = \frac{1}{3}(1 + 1 + 1) = 1$. □

($\mathbf{A}_{\text{Lean}}$; Lean: `gte_baryon_number_topological_charge`, zero sorry.)

8.3 Topological conservation via the Bianchi identity

The baryon current is

$$J_B^\mu(\mathbf{x}) = \frac{7}{6\pi} \sum_j \chi_q(w_j(\mathbf{x}))\, \varepsilon^{\mu\nu} \partial_\nu (\Phi_{\text{MDL}})_j(\mathbf{x}),$$

where $(\Phi_{\text{MDL}})_j$ is the Φ_{MDL} field on tape j. This current is conserved by the Bianchi identity $\varepsilon^{\mu\nu}\partial_\mu\partial_\nu = 0$:

$$\partial_\mu J_B^\mu = \frac{7}{6\pi} \sum_j \chi_q(w_j)\, \varepsilon^{\mu\nu} \partial_\mu \partial_\nu (\Phi_{\text{MDL}})_j = 0.$$

This conservation is automatic—it does not require a Noether current or a continuous symmetry. Baryon number is topological: it counts net kink winding of the Φ_{MDL} field and is conserved identically. (**A/D**; algebraic consequence of Bianchi identity.)

8.4 Verification on SM vertices

All 33 SM charged-current vertices conserve baryon number (**A**, exhaustive computational scan). Six representative vertex types are algebraically certified in Lean. Examples:

- Proton: $B(2, 2, 6) = \frac{1}{3}(1 + 1 + 1) = 1$. ✓
- Photon: $B(0, 0, 0) = 0$. ✓
- Anti-proton: $B(5, 5, 1) = \frac{1}{3}(-1 - 1 - 1) = -1$. ✓
- Quark-gluon vertex $u \to u + g$ ($w = 2 \to 2 + 0$): $\Delta B = 0$. ✓
- Weak vertex $u \to d + W^+$ ($w = 2 \to 6 + 3$): $\Delta B = 1/3 - 1/3 = 0$. ✓

($\mathbf{A}_{\text{Lean}}$ for six representative vertex types; **A** for all 33 by exhaustive scan. Lean: `baryon_number_z7_conserved` (bundle, `BaryonNumber.lean`, `ugp-lean`), zero sorry.)

8.5 Quark identification from fractional charge

The sectors $w \in \{2, 6\}$ are identified as quarks by their electric charge: $Q = w/3 = 2/3$ (up-type) and $Q = 6/3 \equiv -1/3 \pmod{7/3}$ (down-type). This fractional charge identification is algebraic: no additional assumption is made about the quark-hadron duality or the confinement mechanism beyond what is established in Section 9.

9 Color Confinement and SU(3) Structure

9.1 MDL derivation of color confinement

Theorem 9.1 (Color confinement)**.** *The MDL description-length gap between a free colored quark state and a color-neutral hadron state is*

$$\Delta K = \log_2(N_c^2) = \log_2(9) \approx 3.17 \text{ bits.}$$

Free colored quarks are forbidden as asymptotic states because $\Delta K > 0$ makes them non-minimal in the PSC framework.

Proof. A free colored quark of color r on tape x requires specification of: (a) which tape carries the quark winding: $\log_2 N_c = \log_2 3$ bits; (b) which quark sector is occupied: $\log_2 N_c = \log_2 3$ bits. Total: $\log_2 3 + \log_2 3 = \log_2 9$ bits.

A color-neutral hadron (baryon or meson) requires specification of: the winding value from the PSC-admissible set $\mathcal{W}$: $\log_2 |\mathcal{W}| = \log_2 5$ bits per tape.

The description-length gap is $\Delta K = K_{\text{colored}} - K_{\text{neutral}} = \log_2 9 - \log_2 5 = \log_2(9/5) < \log_2 2 =$ 1 bit.

More precisely, the relevant comparison is between the isolated colored configuration $K_{\text{colored}}(w, 0, 0) = \log_2 45$ (specifying which quark winding on which tape, with the other tapes fixed) and the color-neutral configuration $K_{\text{neutral}}(w, w, w) = \log_2 5$ (specifying the shared winding value). The gap is $\Delta K = \log_2 45 - \log_2 5 = \log_2 9 \approx 3.17$ bits.

Since $\Delta K > 0$, free colored states cost more bits than color-neutral ones. The PSC/MDL principle requires physical asymptotic states to be MDL-minimal; therefore free colored quarks are excluded. □

($\mathbf{A}_{\text{Lean}}$; Lean: `color_confinement_k_extra_pos`, zero sorry.)

9.2 $N_c = 3$ from the DPP construction

The number of colors is not an independent assumption:

$$N_c = N_{\text{tapes}} = N_{\text{spatial dims}} = 3.$$

The Dimensional Protocol Principle (DPP) construction of the three-tape architecture fixes three tapes to represent three spatial dimensions. The color index is the tape label. This means that $N_c = 3$ is a structural consequence of 3+1D spacetime, not an independent input.

The boundary value $c_H = 13$ satisfies $2c_H + 1 = 27 = N_{\text{gen}}^3$ (`two_cH_plus_one_eq_ngen_cubed`, zero sorry). This is one manifestation of a deeper structural coincidence: the cascade arithmetic formula for b_{L_1}, $G(n) = n^4 - (n^2+1)/2 - n$, and the Frobenius chain formula $F(n) = n^4 - n^2 + 1$ agree if and only if $n = N_c = 3$, since $F(n) = G(n)$ reduces to $(n-3)(n+1) = 0$. The Frobenius chain generates exactly the two GTE primes $\{7, 73\} = \{|\mathbb{Z}_7|, b_{L_1}\}$, terminating at level 3 ($703 = 19 \times 37$, composite). This Frobenius-Generation Coincidence Identity is machine-certified in Lean 4 (`fgci_unique_at_nc`, `UgpLean.Universality.FrobeniusChain`, 17 theorems, zero sorry).

9.3 SU(3) from the $\Delta w = \pm 1$ selection rule

The elementary CMCA step has a selection rule: only transitions with $\Delta w = \pm 1$ (single-step transitions in $\mathbb{Z}_7$) are elementary. For the quark sectors $w \in \{2, 6\}$ on three tapes, the allowed elementary charge vectors are:

Charge vector $(\Delta w_x, \Delta w_y)$	Gluon type
$(1,0)$, $(6,0)$	Red→Green, Green→Red
$(0,1)$, $(0,6)$	Red→Blue, Blue→Red
$(1,6)$, $(6,1)$	Green→Blue, Blue→Green

These 6 off-diagonal gluon charge vectors, forming two $\mathbb{Z}_3$ orbits, plus 2 Cartan generators from the diagonal (λ_3 and λ_8), give exactly $6 + 2 = 8$ SU(3) generators.

Theorem 9.2 (SU(3) gluon count)**.** *The $\Delta w = \pm 1$ selection rule on three tapes with quark sectors $\{2, 6\}$ gives exactly 8 gluon generators, matching the dimension of SU(3).*

($\mathbf{A}_{\text{Lean}}$; Lean: `su3_gluon_charge_vectors`, `su3_cmca_master_bundle`, zero sorry.)

9.4 Baryon color neutrality

A baryon contains one quark on each tape. The three-tape $\mathbb{Z}_3$ permutation symmetry of the CMCA (tapes related by cyclic permutation $x \to y \to z \to x$) forces the baryon state to be the symmetric superposition

$$|B\rangle = \frac{1}{\sqrt{3}} \left[T + \mathbb{Z}_3(T) + \mathbb{Z}_3^2(T) \right],$$

where T is any single-color tape configuration. This superposition is color-neutral by construction: all three colors appear with equal weight.

($\mathbf{A}_{\text{Lean}}$; Lean: `baryon_color_z3_orbit_neutral`, zero sorry.)

9.5 3D CA rule impossibility

Theorem 9.3 (3D CA rule impossibility ($\mathbf{A}$))**.** *No function $f : \mathrm{GF}(7)^9 \to \mathrm{GF}(7)$ can simultaneously encode Rule 110 dynamics within-axis and $\mathbb{Z}_7$ gravitational coupling across-axis. The three-tape factored architecture is the unique minimal solution.*

This result is established computationally by exhaustive search over the space of candidate 3D CA rules and confirmed by the algebraic argument that the within-tape chirality requirement and the cross-tape symmetry requirement are incompatible for any single lookup function over nine inputs. (**A**; computationally verified.)

10 Electroweak Structure and the W Boson

10.1 Charge from the degree-1 center projection

Theorem 10.1 (Charge from polynomial)**.** *For all PSC-admissible winding numbers $w \in \mathcal{W}$,*

$$3Q(w) = p(0, w, 0) = w \pmod 7,$$

where $Q(w)$ is the electric charge of the particle in units of the proton charge.

Proof. Direct computation: $p(0, w, 0) = w + 0 - 0 \cdot w - 0 \cdot w \cdot 0 = w$. The charge assignment $Q = w/3$ follows from the quark charge $Q_u = 2/3$ requiring $w_u = 2$ and the lepton charge $Q_e = -1$ requiring $w_e = -3 \equiv 4 \pmod 7$. □

($\mathbf{A}_{\text{Lean}}$; Lean: `gte_charge_is_linear_projection`, zero sorry.)

10.2 Gravity/EM degree split

Theorem 10.2 (Gravity/EM degree split)**.** *From the same 19-bit polynomial:*

- Gravity *arises from the degree-3 diagonal* $p(w, w, w)$*: the mass source is cubic in the winding, giving the mass hierarchy.*
- Electromagnetism *arises from the degree-1 center projection* $p(0, w, 0) = w$*: the charge source is linear in the winding.*

Both forces are derived from the same 19-bit description, with zero extra specification cost for either.

($\mathbf{A}_{\text{Lean}}$; Lean: `gravity_em_degree_split`, zero sorry.)

The degree split has a physical consequence: gravitational mass grows as w^3 (giving the observed mass hierarchy among SM particles) while electric charge grows as w^1 (giving the linear charge quantization). These are not coincidences or fine-tunings; they are the degree-3 and degree-1 terms of the same cubic polynomial.

A corollary is that a single-tape source contributes zero gravitational density: $p(w, 0, 0) = 0$ for all w. Gravity requires cross-tape coordination — the cubic $-w_x w_y w_z$ term in p vanishes unless all three tapes are simultaneously non-vacuum. EM charge, being degree-1, requires only a single non-vacuum tape. ($\mathbf{A}_{\text{Lean}}$; Lean: `gravity_requires_cross_tape_coordination`, zero sorry.)

10.3 The universal massless propagator

From Section 6, all massless bosons share the propagator $(-\Delta)^{-1} = 1/(4\pi r)$. This means the graviton propagator, the photon propagator, and the gluon propagator (ultraviolet, UV) are all identical—not by assumption but because they all arise from the same three-tape causal-graph Laplacian. This is the deepest form of gravitational/gauge unification in the GTE framework.

10.4 Lepton-W universality

As noted in Section 4, the identity $-w(e^-) \equiv w(W^+) \pmod 7$ is an arithmetic fact. Its physical consequence is lepton-W universality: all three lepton generations couple to W bosons with equal strength because they all share $w = 4$ and $-4 \equiv 3 = w(W^+) \pmod 7$.

10.5 W bosons as angular modes

The W^+ boson arises as an angular mode of the Φ_{MDL} field:

$$\Delta w(W^+) = (w_u - w_d) \bmod 7 = (2 - 6) \bmod 7 = 3 = w(W^+).$$

The potential $V_{\mathbb{Z}_7}(|\Psi|)$ depends only on $|\Psi|$, making it invariant under the $\mathrm{SU}(2)_L$ transformation $\Psi \to U\Psi$ since $|U\Psi| = |\Psi|$.

The MDL principle forces the gauging of this $\mathrm{SU}(2)_L$ symmetry: a global symmetry costs $K_{\text{extra}} > 0$ bits (specifying the symmetry parameter at every point), while a gauged symmetry costs zero extra bits (the gauge field is the redundancy). This PMDL gauging mechanism introduces the W^a_μ gauge fields with $K_{\text{extra}} = 0$. ($\mathbf{A}_{\text{Lean}}$; zero named axioms; `su2l_l2_from_phimdl_potential_catad` packages `phimdl_potential_su2l_invariant`, `su2l_covariant_derivative_minimal`, and `su2l_wpm_generator_algebra`, all zero `sorry` in `GaugeMDL.lean`.)

10.6 W boson mass

The W boson mass arises from the Higgs mechanism via the SRRG condensate $v_{\mathrm{PSC}} = 246.16$ GeV (Section 11):

$$m_W = \frac{g\, v_{\mathrm{PSC}}}{2} \approx 80.37 \text{ GeV},$$

within 0.003% of the PDG 2024 value $m_W = 80.3692$ GeV, using the 1-loop weak coupling $g = 0.653$ at the electroweak (EW) scale. The vacuum expectation value v_{PSC} is Lean-certified (grade [A$^-$]) from the SRRG fixed-point equation (Section 11). The W mass result is **A/D** (analytic formula with Lean-certified VEV).

11 The SRRG Golden Ratio Bridge

11.1 The self-referential fixed-point equation

The vacuum fixed-point equation $p(x, x, x) \equiv x \pmod 7$ was shown in Theorem 2.3 to have no solution in $\mathbb{Z}_7$ other than $x = 0$. Over $\mathbb{R}$, however, the same polynomial equation $2x - x^2 - x^3 = x$, i.e., $x - x^2 - x^3 = 0$, factors as $x(1 - x - x^2) = 0$, giving $x = 0$ or $x^2 + x - 1 = 0$.

The positive real root of $x^2 + x - 1 = 0$ is

$$x_+ = \frac{-1 + \sqrt{5}}{2} = \frac{1}{\phi} = 0.6180\ldots,$$

the reciprocal of the golden ratio $\phi = (1 + \sqrt{5})/2$.

Theorem 11.1 (SRRG-CA bridge)**.** *The positive real root of the CA self-referential fixed-point equation $p(x, x, x) = x$ is $1/\phi = (\sqrt{5} - 1)/2$. This equals the SRRG contraction eigenvalue.*

(**A**$_{\mathrm{Lean}}$; Lean: `gte_poly_srrg_bridge`, zero sorry.)

11.2 Physical meaning

The SRRG is the renormalization group flow associated with the Φ_{MDL} field's self-referential structure. The contraction eigenvalue $\lambda_{\mathrm{SRRG}} = 1/\phi$ governs the RG flow in the self-similar fixed-point regime. The fact that this eigenvalue is the positive root of the CA polynomial's own fixed-point equation means that the renormalization group structure of Φ_{MDL} is encoded in the polynomial p itself: the golden ratio is not an additional input but an algebraic consequence of the 19-bit specification.

11.3 The Higgs VEV from SRRG

The electroweak VEV v_H is derived from the SRRG condensate. The first-principles formula gives

$$v_H = 14\pi^2 m_\tau \approx 245.52 \text{ GeV} \quad (-0.29\% \text{ from PDG } 246.22 \text{ GeV}),$$

and the exact SRRG fixed-point result (Lean-certified, grade [A$^-$]) gives

$$v_{\mathrm{PSC}} = 246.16 \text{ GeV} \quad (-0.024\% \text{ from PDG } 246.22 \text{ GeV}).$$

This is the VEV used in the W mass calculation (Section 10.6). The Lean certification is `srrg_higgs_vev_from_fixed_point` (alias: `ew_vacuum_bridge_grade_certificate`; module: `srrg-lean/VEVProof/EWVacuumBridge.lean`; zero sorry; conditional on `PhysicalSubspace` axioms encoding Landauer sustainability and IR-stability); grade [A$^-$].

Remark 11.2. The golden ratio appears in the GTE framework not as a numerological coincidence but as the self-similar scale at which the CA polynomial's real fixed-point equation is satisfied. The SRRG derivation connects this algebraic fact to the observable EW scale.

12 Fermionic Statistics and the Braid Atlas

12.1 Algebraic fingerprint of fermionic sectors

The exchange statistics of a particle are determined by the order of its winding number in $\mathbb{Z}_7^*$. From Section 4:

- $w \in \{2, 4, 6\}$: non-primitive roots of $\mathbb{Z}_7^*$ (orders 3, 3, 2); these are the fermionic sectors.
- $w \in \{3\}$: primitive root (order 6, generator of $\mathbb{Z}_7^*$); this is the bosonic weak-current sector.

The exchange phase formula at Level 1 (algebraic shadow) assigns:

$$\phi_{\text{exchange}}(w) = -1 \quad \text{for} \quad w \in \{2, 4, 6\} \quad \text{(fermionic)}, \tag{11}$$

$$\phi_{\text{exchange}}(w) = +1 \quad \text{for} \quad w \in \{0, 3\} \quad \text{(bosonic)}. \tag{12}$$

Theorem 12.1 (Fermionic sector identification)**.** *The PSC-admissible winding sectors that carry fermionic exchange statistics are exactly the non-primitive roots of $\mathbb{Z}_7^*$.*

($\mathbf{A}_{\text{Lean}}$; Lean: `gte_winding_identifies_charged_fermions`, zero sorry.)

12.2 The Braid Atlas and Level 2 spin-statistics

The full spin-statistics theorem at Level 3 requires identifying the exchange phase $e^{2\pi i s}$ (where s is the spin) with the algebraic exchange phase $\phi_{\text{exchange}}(w)$. This identification is established at Level 1 by the algebraic fingerprint above. The full Level 3 Braid Atlas construction connects this to $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}/2\mathbb{Z}$ and the spinor representation.

One axiom remains named in the Lean formalization: `spinor_exchange_equals_2pi_rotation`, corresponding to the identification of the $\mathbb{Z}/2\mathbb{Z}$ algebraic exchange phase with the physical 2π rotation phase. This will be completed when the Lorentzian extension of the CMCA path integral (OQ-QG-1-Z_2-EFT) is established.

12.3 Gorard coarse-graining coefficient

The Gorard coarse-graining coefficient C_{Gorard} relates the causal-graph Ricci curvature to the energy-momentum tensor through the Ollivier-Ricci scalar. For the three-tape CMCA in the vacuum (Ricci-flat) regime:

$$C_{\text{Gorard}} = \frac{\kappa_{3D}}{8\pi} = 0.0923,$$

where κ_{3D} is the three-dimensional Ollivier curvature normalization. This is an exact algebraic identity, not a fit. ($\mathbf{A}_{\text{Lean}}$; Lean: `three_tape_gorard_vacuum_ricci_flat`, zero sorry.)

13 Falsifiable Predictions

The GTE polynomial framework makes the following zero-parameter predictions:

Quantity	GTE prediction	Measurement	Status
Cosmological constant Ω_Λ	0.6899 (zero param.)	0.6889 ± 0.0056 (Planck 2018)	$+0.18\sigma$
Number of colors N_c	$N_c = N_{\text{tapes}} = 3$	$N_c = 3$ (established)	Structural (DPP)
Baryon number per quark	$B = 1/N_{\text{tapes}} = 1/3$	$B = 1/3$ (established)	Topological, Bianchi
Bell violation S	$S = 2.4459$ at $G_{\text{eff}} = 0.5$	$S \leq 2\sqrt{2}$ (QM bound)	Verified; 86.5% Tsirelson; LHV excluded
Newtonian force law	Eq. (5)	$F \propto b^{-2}$ (Newton)	**A/D**; exact erf formula
W boson mass m_W	80.37 GeV (1-loop $g = 0.653$)	80.3692 ± 0.0133 GeV	0.003% error
SRRG contraction eigenvalue	$1/\phi = 0.6180$	(structural)	Lean-certified algebraic
ΔK (confinement)	$\log_2 9 \approx 3.17$ bits	confinement observed	Lean MDL theorem
CMB tilt n_s	$1 - \ln 2/(2\pi^2) = 0.9649$	0.9649 ± 0.0042 (Planck)	-0.005σ (Lean-certified; physical $\mathbb{Z}_2$-EFT interpretation open)

Falsification criteria.

- Ω_Λ: a measurement establishing $\Omega_\Lambda < 3\pi/14 = 0.6732$ (below the carrier floor) or $\Omega_\Lambda > 0.6899$ (above the census ceiling) at $> 3\sigma$ would falsify the PMDL bracket structure (Section 5.4).
- N_c: any precision experiment finding $N_c \neq 3$ would falsify the DPP three-tape construction.
- $B = 1/3$: any violation of baryon number conservation (decay $p \to e^+\pi^0$, etc.) would falsify the topological conservation mechanism.
- m_W: a measured value deviating from $g\, v_{\text{PSC}}/2 = 80.37$ GeV by more than 1% after 2-loop corrections would require re-examination of the SRRG VEV.

The structural meta-prediction. The deepest prediction of the GTE framework is architectural: all massless bosons (graviton, photon, gluon) share the same propagator $(-\Delta)^{-1}$. This is testable in principle through precision measurements of dispersion relations at high energy: any deviation in the UV propagator of gravity relative to EM would falsify the universal Laplacian identification.

14 Limitations and Open Problems

14.1 String tension and Level 2 QCD

The linear confinement potential $V(r) \sim \sigma r$ (string tension) is a Level 2 QCD phenomenon. The MDL confinement theorem of Section 9 establishes that free colored quarks are non-minimal and therefore forbidden, but does not derive the string tension σ explicitly. This requires the Level 2 QCD sector construction, which is the subject of ongoing work.

14.2 Baryon number at Level 2

The topological baryon number established in Section 8 is a Level 1 result: it counts net Φ_{MDL} kink winding with the quark indicator χ_q. The Level 1 and Level 2 baryon numbers coincide (**A/D**):

$$B_{L2} = \int J_B^0 \, d^2x = \tfrac{1}{3} \sum_j \chi_q(w_j) = B_{L1}$$

via the exact algebraic identity $(7/6\pi)(2\pi/7) = 1/3$. Conservation $\partial_\mu J_B^\mu = 0$ holds topologically from $\varepsilon^{\mu\nu}$ antisymmetry and the Schwarz lemma. Lean certificate: `baryon_number_l1_l2_correspondence_certified` (zero sorry, `ugp-lean/UgpLean/Algebra/BaryonNumber.lean`). (*Script:* `papers/46_gte_polynomial_uft/scripts/baryon_number_l2_correspondence.py`.)

14.3 CMB spectral tilt

The prediction $n_s = 1 - \ln 2/(2\pi^2) = 0.96488$ (at -0.005σ from Planck 2018) is machine-certified in Lean 4 (14 zero-sorry theorems, zero axioms; `cmca_z2_sublayer_spectral_tilt` and related; `UgpLean.Gravity.CMBSpectralTilt`; $\mathbf{A}_{\mathrm{Lean}}$), including the definitional discharge of the $\mathbb{Z}_2$-EFT bridge. The physical EFE-bridge interpretation of that step remains the residual derivation gap; the full derivation-chain status is assessed in the companion paper [5].

14.4 Lorentzian extension

The three-tape CMCA operates in Euclidean time (causal period τ_c). The full Lorentzian extension requires an SO(1,3) Lorentz group representation on the $\mathbb{Z}_7$ winding sectors. The Gorard coarse-graining theorem provides the Euclidean Ricci-flat vacuum; the Lorentzian extension is blocked by the absence of a Lean library for SO(1,3) in the current Mathlib. When this becomes available, the spectral dimension calculation will follow.

14.5 Open question register

Open Problem 14.1 (Z_2-EFT coupling (OQ-QG-1)). Establish the physical EFE-bridge interpretation of the Z_2-EFT coupling that connects the CMCA binary gate entropy $\ln 2$ to the gravitational effective action via the Wald entropy formula (the Lean side is definitionally discharged; Section 14.3). Closing it would also provide the Level 3 completion of the spin-statistics theorem.

Open Problem 14.2 (Domain selection (Conjecture C2)). Prove the uniqueness of the physical domain $D \in [\mathbf{D}]$ within the equivalence class of Φ_{MDL} completions. This is the long-term programme of the GTE uniqueness agenda (see [4] §10.2).

15 Discussion and Conclusion

15.1 Summary of results

We have established that a single polynomial $p(L, C, R) = C + R - CR - LCR \pmod 7$, requiring $K_{\mathrm{CMCA}} = 19$ bits to specify, achieves MDL unification of five independent physical domains with $K_{\mathrm{extra}} = 0$ for each:

(1) **Spatial dynamics:** Rule 110 Turing universality on binary restriction.

(2) **Gauge coupling:** $\mathbb{Z}_7$ winding conservation at all 33 SM vertices.

(3) **Gravity:** PMDL variational equation $\nabla^2\Phi = G_{\text{eff}}\, p$.

(4) **Quantum entanglement:** Negativity $\mathcal{N} = 0.24$; Bell $S = 2.4459$ (86.5% Tsirelson); LHV excluded; Born rule from τ_c PW clock (**A/D**).

(5) **Baryon number:** Topological charge $B = (1/3)\sum_j \chi_q(w_j)$.

From these five roles, the following are derived with no additional assumptions: the number of colors $N_c = 3$ (from $N_{\text{tapes}} = 3$); the cosmological constant $\Omega_\Lambda = 0.6899$ ($+0.18\sigma$ from Planck 2018); the W boson mass $m_W = 80.37$ GeV; the golden ratio as the SRRG contraction eigenvalue; color confinement with description-length gap $\Delta K = \log_2 9$; and the fermion/ boson split from the order structure of $\mathbb{Z}_7^*$.

The generating functional $Z[J]$ provides a unified functional framework in which gravity, electromagnetism, and the strong force all arise from the same $(-\Delta)^{-1}$ propagator, with the degree of the polynomial source distinguishing the three forces.

15.2 Relation to existing unification programmes

The GTE unification differs from conventional grand unified theories (GUT) in a fundamental way. GUTs embed the SM gauge group into a larger group and then break it; this requires specifying the larger group, the breaking pattern, and the Higgs sector—all additional specification costs. The UGP Physics programme instead derives all structure from a minimal 19-bit description, with zero extra specification cost for each physical role.

The MDL principle as a physical selector has been proposed in various forms (Kolmogorov complexity, minimum description length in statistical learning). The GTE framework instantiates this principle at the level of fundamental physics: the physical laws are the shortest program for the observable universe.

15.3 What P46 establishes and what remains

Lean-certified ($\mathbf{A}_{\text{Lean}}$, zero sorry): All foundational algebraic results—vacuum uniqueness, MDL uniqueness, charge formula, gravity/EM degree split, SU(3) gluon count, color confinement, baryon number, fermion/boson split, SRRG bridge, PSC dark energy.
Analytically derived (A/D): PMDL-Λ duality; $Z[J]$ unified generating functional; W angular mode mechanism; $\Omega_\Lambda = 0.6899$; exact Newtonian force law $\varphi(b) = G_{\text{eff}} M\, \text{erf}(b/\sqrt{2}\sigma_{\text{AL}})/(4\pi b)$ with three mechanism levels; Born rule $P(k|\tau) = |\langle k|U(\tau)|\psi_0\rangle|^2$ from τ_c Page–Wootters clock.
Computationally verified (A): Universal massless propagator; 3D CA rule impossibility; Bell $S = 2.4459$ (86.5% Tsirelson; LHV excluded); entanglement negativity $\mathcal{N} = 0.24$.
Open: String tension at Level 2; Lorentzian extension; physical interpretation of the Z_2-EFT bridge in the CMB tilt chain.

15.4 Conclusion

The GTE polynomial $p(L, C, R) = C + R - CR - LCR \pmod 7$ is the simplest object in this paper, requiring 19 bits to specify. It is also—as this paper demonstrates—sufficient to certify the existence of gravity (with exact Newtonian force law $F \propto b^{-2}$), gauge interactions, quantum entanglement

(Bell $S = 2.4459$, LHV excluded), the Born rule (from τ_c Page–Wootters clock), baryon number conservation, and Turing-complete computation within a single unified framework.

The companion paper [7] establishes this as the **Single-Source Principle**: the same 19-bit polynomial that determines the spatial dynamics of one CMCA tape simultaneously governs gauge coupling, gravity, entanglement, and baryon number, with zero extra specification cost for each role. The present paper provides the full algebraic derivation confirming that the single-source structure is not an architectural coincidence but a provable algebraic consequence of MDL minimality over GF(7).

The MDL principle selects p uniquely among all cubic polynomials over GF(7) satisfying the mass hierarchy constraints. The three-tape architecture fixes the number of colors, the baryon charge unit, and the universal propagator without additional input. The PMDL variational principle derives both gravity and the cosmological constant from the same action.

In this sense, 19 bits is the answer to the question "what is the minimum specification of a unified field theory?"

The algebraic structure of p as a standalone $\mathbb{Z}_7$ dynamical system — including the invariant subset classification, the QNR binary-floor theorem explaining why $\{0, 1\}$ is the unique proper invariant subset, and the Schwartz–Zippel proof that f_{MDL} is algebraically distinct from p in kind — is studied in the companion paper P49 [3], which also provides the Wolfram Physics Project framing of the MDL selection result.

The arithmetic depth of $N_c = 3$ extends beyond the three-tape architecture: the Frobenius prime chain $F(n) = n^4 - n^2 + 1$ at $N_c = 3$ generates exactly $\{7, 73\} = \{|\mathbb{Z}_7|, b_{L_1}\}$ and terminates ($703 = 19 \times 37$ is composite), providing two independent derivations of $b_{L_1} = 73$ that agree uniquely at $n = N_c = 3$ (Frobenius-Generation Coincidence Identity; `fgci_unique_at_nc`, `UgpLean.Universality.FrobeniusChain`, 17 theorems, zero `sorry`). The single integer $N_c = 3$ thus determines the field, the seed, and the tape count simultaneously.

A Lean Certification Inventory

All theorems are in the `ugp-lean` repository. The modules listed below have zero `sorry`.

Theorem name	**Module**	**Physical role**
`gte_polynomial_five_roles_k_extra_zero`	`FiveRolesPolynomial.lean`	Single-Source Principle: five roles at $K_{\mathrm{extra}} = 0$
`unique_cubic_gravity_coupling`	`PMDLGravityTheorems.lean`	MDL uniqueness of p
`vacuum_unique_fixed_point_z7`	`PMDLGravityTheorems.lean`	Vacuum $w = 0$ unique fixed point
`gte_charge_is_linear_projection`	`ChargeFromPolynomial.lean`	$3Q(w) = p(0, w, 0) = w$
`gravity_em_degree_split`	`ChargeFromPolynomial.lean`	Gravity/EM degree split
`su3_gluon_charge_vectors`	`SU3GluonCount.lean`	8 SU(3) gluon generators
`su3_cmca_master_bundle`	`SU3GluonCount.lean`	SU(3) from $\Delta w = \pm 1$
`color_confinement_k_extra_pos`	`ColorConfinementMDL.lean`	$\Delta K = \log_2 9 > 0$

Theorem name	Module	Physical role
gte_baryon_number_ topological_charge	BaryonNumber.lean	$B = (1/3)\sum \chi_q$
baryon_number_z7_conserved	BaryonNumber.lean	Baryon conservation (Bianchi)
baryon_color_z3_orbit_ neutral	SU3GluonCount.lean	Baryon color-neutrality
baryon_number_l1_l2_ correspondence_certified	BaryonNumber.lean	$B_{L2} = B_{L1}$ via $(7/6\pi)(2\pi/7) = 1/3$; $\partial_\mu J^\mu_B = 0$ topological (**A/D**)
gte_winding_identifies_ charged_fermions	WindingToBraidRep.lean	Fermion/boson $\mathbb{Z}_7^*$ split
gte_poly_srrg_bridge	SRRGCABridge.lean	$1/\phi$ from CA fixed point
srrg_higgs_vev_from_fixed_ point	EWVacuumBridge.lean (srrg-lean)	$v_{\text{PSC}} = 246.16$ GeV (-0.024%); grade [A$^-$]; see §11.1
no_final_self_theory	PSCEpochSelection.lean	$D_{\text{res}} > 0$ (dark energy)
psc_undecidability_residual_ pos	PSCEpochSelection.lean	$D_{\text{res}} > 0$
d_res_determines_omega_ lambda	PSCEpochSelection.lean	Ω_Λ uniquely fixed
psp_epoch_selection_master	PSCEpochSelection.lean	Master bundle: D_{res}, Ω_Λ
incompleteness_implies_ nonzero_omega_lambda	PSCEpochSelection.lean	PSC incompleteness $\Rightarrow$ $D_{\text{res}} > 0 \Rightarrow \Omega_\Lambda > 0$ ($\mathbf{A}_{\text{Lean}}$)
mdl_tower_bundle	MDLTower.lean	Three nested MDL roles unified ($\mathbf{A}_{\text{Lean}}$)
three_tape_gorard_vacuum_ ricci_flat	GorardRicciFlatVacuum.lean	$C_{\text{Gorard}} = \kappa_{3D}/(8\pi)$
three_tape_causal_diamond_t4	GorardRicciFlatVacuum.lean	$V(T) = T^4/4$; $\text{Vol}(S^3) = 2\pi^2$
tau_c_pw_clock_validity	(pending Mathlib PW lib.)	τ_c PW clock prerequisites; Born rule $P(k\|\tau) = \|\langle k\|U\|\psi_0\rangle\|^2$ (**A/D**; Lean cert pending)
rule110_z7_poly_rep	Univ./PhiMDL-Universality.lean	p = Rule 110 on GF(7)
rule110_center1_is_nand	Univ./PhiMDL-Universality.lean	$\ln 2$ from Z_2 binary gate
gte_winding_sm_vertex_ conserved	Univ./GUTStructure.lean	6 representative SM vertices conserve $\mathbb{Z}_7$ ($\mathbf{A}_{\text{Lean}}$); all 33 by exhaustive scan (**A**)
gte_winding_sm_vertex_ conserved_full	Univ./GUTStructure.lean	All 33 SM vertex types conserve $\mathbb{Z}_7$ ($\mathbf{A}_{\text{Lean}}$, `decide`)

Theorem name	Module	Physical role
phimdl_potential_su2l_invariant	GaugeMDL.lean	L_2 norm preserved under $SU(2)$ rotation ($\mathbf{A}_{\text{Lean}}$)
su2l_covariant_derivative_minimal	GaugeMDL.lean	Covariant derivative MDL-minimal ($\mathbf{A}_{\text{Lean}}$)
su2l_wpm_generator_algebra	GaugeMDL.lean	$W^{\pm}$ satisfy $SU(2)$ Lie algebra ($\mathbf{A}_{\text{Lean}}$)
su2l_l2_from_phimdl_potential_catad	GaugeMDL.lean	Master bundle: full $SU(2)_L$ gauging chain ($\mathbf{A}_{\text{Lean}}$)

Table 2: Lean certification inventory. All modules in `ugp-lean` repository. All theorem sorry counts: 0.

Named axioms (not sorry; explicitly named pending formalization):

Axiom name	Module	Status
spinor_exchange_equals_2pi_rotation	FermionicStatistics.lean	Pending $\pi_1(\text{SO}(3))$ Lean formalization

B Numerical Verification

All computationally verified results use scripts in `papers/46_gte_polynomial_uft/scripts/`.

Quantity	Value	Error	Script / method
Ω_Λ	0.6899	$+0.18\sigma$ from Planck	`PSCEpochSelection.lean` (analytical bound)
ΔK (confinement)	$\log_2 9 = 3.170$ bits	exact	`ColorConfinementMDL.lean`
Gravity force law	Eq. (5); $b^{-2.004}$ at $b/\sigma = 100$	b^{-2} (Newton)	`gravity_force_law_continuum_limit.py`; **A/D**
Bell violation S	2.4459	86.5% Tsirelson	`bell_inequality_test.py`; $G_{\text{eff}} = 0.5$; LHV excluded
Entanglement negativity	$\mathcal{N} = 0.24$	—	`bell_inequality_test.py`; $G_{\text{eff}} = 0.5$
Born rule (τ_c clock)	Born prob. $P(k\|\tau)$	—	`pw_born_rule_verification.py`; **A/D**
SRRG bridge	$1/\phi = 0.6180$	exact algebraic	`SRRGCABridge.lean`
m_W (1-loop)	80.37 GeV	0.003% from PDG 2024	SRRG VEV $v_{\text{PSC}} = 246.16$ GeV (-0.024%); $g = 0.653$; grade [A$^-$]
SM vertex conservation	33/33	100%	Exhaustive scan; `WindingToBraidRep.lean`
Gorard coefficient	0.0923	exact	`GorardRicciFlatVacuum.lean`

Computational Tools

Numerical computations use Python 3.9+ with NumPy for linear algebra and array operations, and SciPy for filtering and integration.

The machine-checked formalisation is written in Lean 4 with Mathlib. All theorems cited as $\mathbf{A}_{\text{Lean}}$ carry zero `sorry` and zero custom axioms beyond Lean's standard foundations; named physics axioms are explicitly disclosed wherever used.

Full reproduction instructions are provided in `REPRODUCE.md` co-located with this paper's source.

AI coding assistants were used for LaTeX formatting, coding assistance, and adversarial review during manuscript preparation. All mathematical derivations, numerical computations, and Lean 4 certifications were independently verified by deterministic scripts and the Lean 4 kernel.

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